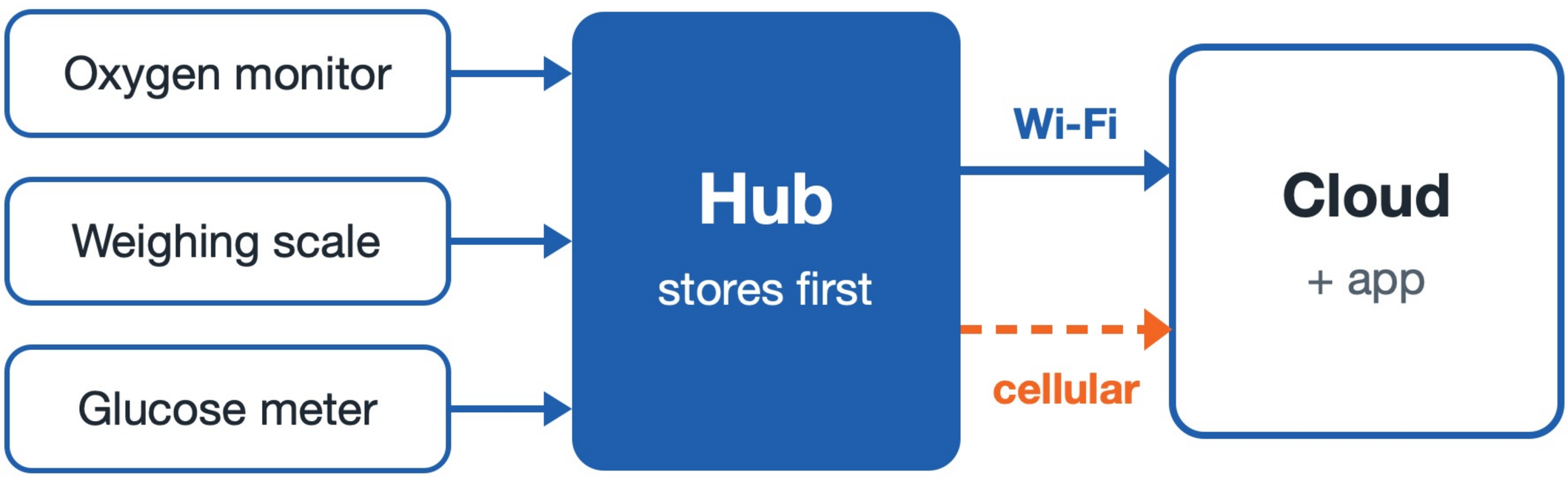
LESSON FROM A HOME HEALTH HUB

# The patient won't fix the Wi-Fi.

So the hub has to look after itself.




Swipe for five design lessons ->

THE PROJECT

# A home hub for chronic care.

A healthcare client in Australia wanted one device that gathers a patient's readings at home and shares them.

**IT COLLECTS**

Oxygen level

Weight

Blood sugar

**WE BUILT**

Product architecture

Hardware, firmware

Linux drivers

Mobile app

The app is designed for elderly patients.

1 CONNECTIVITY

# Wi-Fi first.
# Cellular as backup.

At home, nobody owns the network.
The router gets swapped, the password changes, the power goes off at night.

| | | |
|---|---|---|
| Wi-Fi works | → | **send over Wi-Fi** |
| Wi-Fi drops | → | **switch to cellular** |
| Wi-Fi returns | → | **move back to Wi-Fi** |

For a medical hub, a dropped link is a patient-safety issue.

2 DATA

# Store first.
# Send second.

Every reading is saved on the hub before it goes anywhere. When the connection returns, the hub syncs.



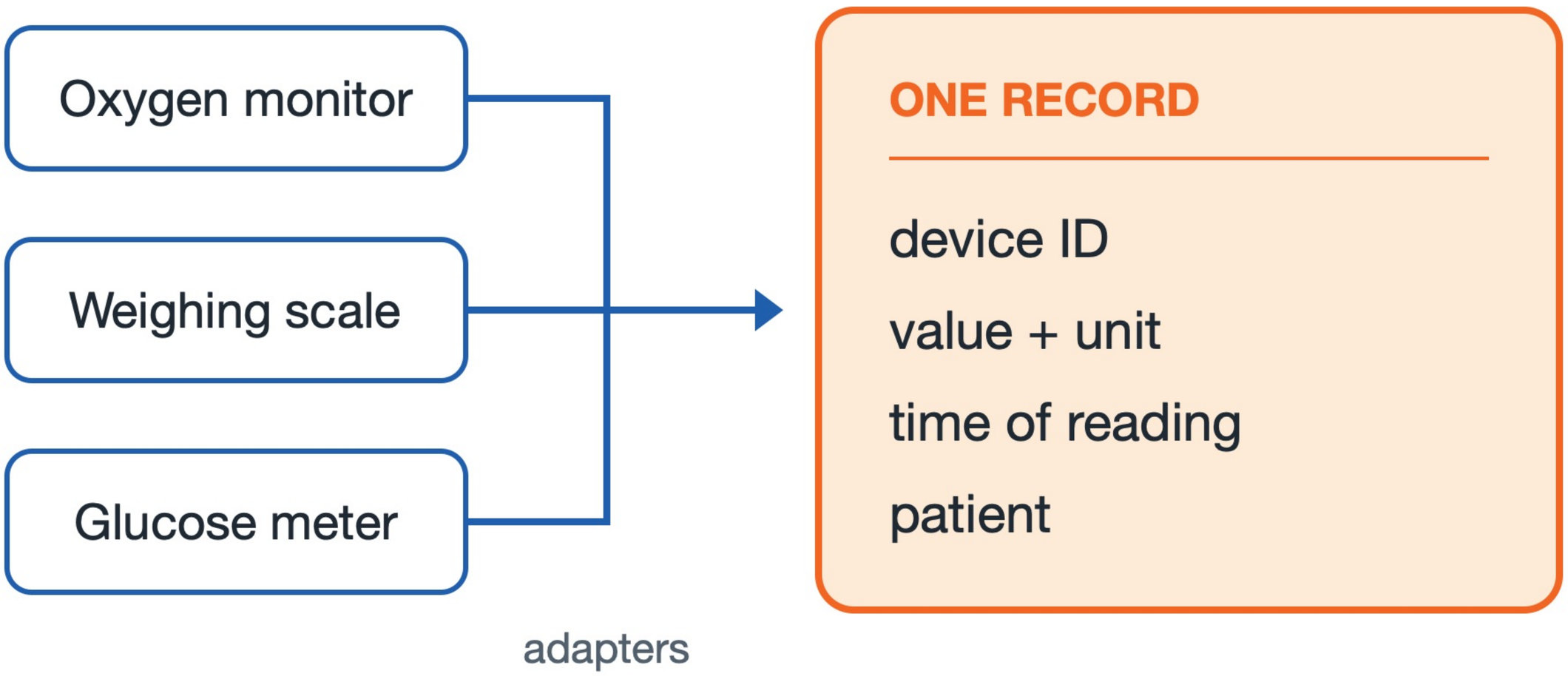

**GIVE EVERY READING AN ID**

A retry then updates one record.
It never creates a duplicate.

3 DEVICES

# One record for every device.

Each monitor sends data its own way.
The hub turns all of it into one format.




The cloud and the app only see one format.

4 NO IT DEPARTMENT

# Recover without a home visit.

Nobody will log in to fix it.
So the hub has to:

- Boot back up after a power cut
- Keep its data through power loss
- Restart a stuck service (watchdog)
- Show what's wrong with one light

**ON EMBEDDED LINUX**

A read-only root file system and a journaled data partition help here.

5 CELLULAR BUDGET

# Cellular data costs money.

The backup link carries what matters right now. Everything else can wait.

**OVER CELLULAR**

- Readings
- Alerts
- Remote commands

**WAIT FOR WI-FI**

- Debug logs
- Large updates
- Bulk diagnostics

Decide this in the design, not after the first data bill.

TAKEAWAY

# Design for the home, not the lab.

1. Wi-Fi first, cellular as backup
2. Store first, send second
3. One record for every device
4. Recover without a home visit
5. Cellular for what matters now

A clinician should never have to ask,
"Did the hub miss something?"

**Follow CoreFragment for more**